

PARK-LDNet: A Hybrid LSTM-DBN Framework for Parkinson's Disease Prediction

Dimpul Durga P

Computing Technologies

SRM Institute of Science and Technology
Chennai, India

dp7793@srmist.edu.in

Charmi Akshita K

Computing Technologies

SRM Institute of Science and Technology
Chennai, India

ck9799@srmist.edu.in

Raghu T

Computing Technologies

SRM Institute of Science and Technology
Chennai, India

rt5150@srmist.edu.in

Sailaja K

Computing Technologies

SRM Institute of Science and Technology
Chennai, India

ks3280@srmist.edu.in

Hariharan R

Computing Technologies

SRM Institute of Science and Technology
Chennai, India

hariharr2@srmist.edu.in

Abstract—This paper presents a hybrid deep learning approach for Parkinson's Disease detection using voice biomarkers. By integrating Long Short-Term Memory (LSTM) networks for temporal pattern recognition and Deep Belief Networks (DBN) for hierarchical feature extraction. In this research on classifying the voice features in Parkinson's disease, robust baseline results were achieved utilizing machine learning techniques. The model further incorporated LSTM networks that could capture temporal dependencies in sequential voice data, thus enabling the detection of subtle changes in vocal patterns over time. The model was also implemented with DBN for hierarchical feature extraction to learn complex non-linear relations among voice features. The proposed approach using LSTM and DBN models outperformed the standard machine learning techniques, hence showing a promising result toward early detection of Parkinson's disease. This hybrid framework improves diagnostic accuracy and gives an overall analysis of voice patterns that may serve as a potentially valuable tool in early intervention and patient monitoring. Code will be available at '<https://github.com>'.

Index Terms—Parkinson's Disease, Voice Biomarkers, Machine Learning, Deep Learning, LSTM, DBN, Dimensionality Reduction, PCA, Classification Accuracy, Early Detection, Non-invasive Diagnosis

I. INTRODUCTION

Parkinson's disease is a gradually worsening neurological condition that impacts motor skills, speech, and quality of life, and affects millions of people around the world. Characterized by symptoms like tremors, stiffness, and slowness of movement, PD also affects the vocal attributes, making speech analysis a useful tool in early detection. However, early detection remains elusive because symptoms in the early stages of these diseases are often too subtle to detect and often depend on subjective clinical experience. Standard approaches to diagnosis, including physical examinations and interviews with patients, often miss early signs and therefore delay diagnosis and, by implication, treatment.

Machine Learning and Deep Learning came in, showing that one can indeed develop prediction models automatically by the analysis of some patterns of data for the diagnostic preciseness of Parkinson's disease at earlier stages. Those data patterns may involve voice features that show neurological changes associated with characteristics of jitter and shimmer, along with harmonic to noise ratios. Identification of these minute changes by ML and DL models improves capturing of early signs of PD, hence allowing earlier interventions.

This research enhances the existing machine learning methodologies with respect to incorporating deep learning architectures, namely, LSTM and DBN. The application of the LSTM model best fits the time-series data handling for the analysis of temporal dependencies of voice recordings. This might show the trends of features such as jitter and shimmer over time, hence highlighting the patterns related to the progression of the disease. On the other hand, the DBN will model non-linear relationships in the voice data, hence allowing it to find complex patterns not easily picked up by other traditional statistical analyses. Both models, in conjunction, provide a very strong framework for the early detection of Parkinson's disease with high accuracy, which will lead to better outcomes and an improved quality of life for patients since it aids in earlier and more objective diagnoses.

A. Motivation

The motivation for this work arises from the pressing need for improved diagnostic tools for Parkinson's disease that are non-invasive, accessible, and highly accurate. Voice patterns, being a key indicator of early Parkinson's symptoms, offer a promising avenue for advanced analysis using machine learning and deep learning models. LSTM networks, being well-suited for sequential data, can effectively capture temporal patterns in voice, while DBNs have the potential to uncover intricate relationships among features, further enhancing di-

agnostic accuracy. The objective is to design a model that can detect early changes associated with Parkinson's disease, enabling timely medical intervention.

B. Contribution

This work introduces an innovative integration of machine learning and deep learning approaches to enhance the detection of Parkinson's disease using voice data. The primary contributions include:

- **Temporal Modeling with LSTM:** The adoption of LSTM networks enables the effective modeling of the temporal evolution of voice features, allowing the identification of subtle changes in speech patterns over time for improved prediction accuracy.
- **Hierarchical Feature Learning with DBN:** By leveraging DBNs, the approach achieves robust hierarchical feature extraction, capturing complex and non-linear relationships within voice data. This enhances the model's ability to distinguish between healthy individuals and those affected by Parkinson's disease.
- **Hybrid Model Performance:** The proposed hybrid framework combines the strengths of LSTM and DBN, outperforming conventional machine learning techniques. It not only enhances diagnostic accuracy but also offers a detailed analysis of voice patterns essential for early detection and monitoring of Parkinson's disease.

This research provides a novel diagnostic solution, leveraging the combined strengths of machine learning and deep learning, to develop a non-invasive and effective tool for Parkinson's disease diagnosis.

II. LITERATURE SURVEY

Several studies have explored machine learning and deep learning techniques for Parkinson's disease prediction, highlighting notable limitations. Dr. C.K. Gomathy et al. used SVM and Decision Trees for classification based on voice features but lacked temporal modeling to capture progressive changes [1]. Caudhury and Ghosh worked on a limited dataset, affecting long-term predictions [2]. Yadav and Kumar utilized LSTM for feature extraction but did not address vanishing gradients or compare with traditional models [3]. Ramesh et al. focused on LSTM but neglected DBNs, limiting hierarchical feature learning [4]. Patil and Patil used DBN but overlooked temporal data, missing critical voice patterns [5]. Ayman and Hussein successfully implemented LSTM but lacked comprehensive evaluation metrics [6]. Khemchandani and Sharan emphasized machine learning but ignored LSTM and DBN for improved accuracy [7]. El-Khoury and Delecluse implemented RBMs without stacking layers into DBNs, reducing feature-learning capacity [8]. Lastly, Ghosh and Dutta evaluated a single model type without comparing LSTM and DBN [9]. These studies reveal gaps addressed by integrating LSTM and DBN. Ramamoorthy et al. proposed a LeNet-like architecture for hand sign recognition, demonstrating CNNs' ability to extract spatial features [10]. Inspired by such approaches, our

study introduces a hybrid LSTM-DBN model, which effectively captures both temporal patterns and deep hierarchical features, leading to improved predictive performance.

Existing research in AI-based Parkinson's Disease detection mainly focuses on traditional ML models like SVM, Decision Trees, and CNNs, which struggle with sequential dependencies in voice biomarkers. Our study addresses this gap by introducing a hybrid LSTM-DBN model, which effectively captures both temporal patterns and deep hierarchical features, leading to improved predictive performance.

III. METHODOLOGY

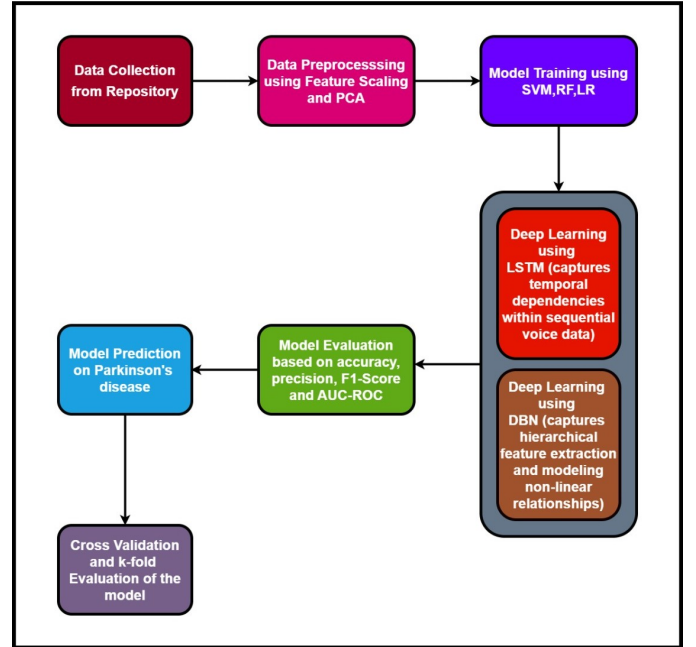


Fig. 1. System Architecture Diagram

A. Data Collection

This project uses the Kaggle data set and focuses on voice records of people with and without Parkinson's disease. The total number of instances in this dataset is 195, and each instance contains 23 features. The target variable "status" indicates whether a person has Parkinson's disease.

The features are derived primarily from voice measurements, which are known to be important biomarkers of Parkinson's disease. Below are some key features used:

TABLE: PARAMETERS AND DESCRIPTIONS

B. Data Preprocessing

1. **Feature Scaling** Feature scaling is crucial to normalize the data, ensuring that each feature contributes equally to the learning process. MinMaxScaler from the sklearn.preprocessing library was applied to scale the features

Parameter	Description
MDVP:Fo(Hz)	Average vocal fundamental frequency.
MDVP:Fhi(Hz)	Maximum vocal fundamental frequency.
MDVP:Flo(Hz)	Minimum vocal fundamental frequency.
Jitter(%)	Variability in frequency.
MDVP:Shimmer	Variability in amplitude.
HNR	Harmonic-to-noise ratio, indicating voice signal quality.
RPDE	Recurrence Period Density Entropy, a measure of signal complexity.
DFA	Detrended Fluctuation Analysis, capturing long-term signal correlations.
PPE	Pitch Period Entropy, indicating signal randomness.

TABLE I
TABLE SHOWING PARAMETERS AND THEIR DESCRIPTIONS.

in the range of $[-1, 1]$. The formula used for MinMax scaling is:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \times (m - n) + n \quad (1)$$

where: m is $\max_{desired}$, n is $\min_{desired}$, X is the original feature value, X_{min} and X_{max} are minimum and maximum values, and X' is the scaled value. max 2. Principal Component Analysis (PCA) To reduce the dimensionality of the dataset while retaining most of the variance, Principal Component Analysis (PCA) was applied, a technique that transforms the dataset into a set of orthogonal components. The goal was to retain 95% of the variance, reducing the original 22 features to 3 principal components. The steps involved in PCA are:

a. Compute the covariance matrix:

$$\text{Cov}(X) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T \quad (2)$$

where: X_i represents the data points, and $\bar{X}$ is the mean of the data set.

b. Eigenvalues and Eigenvectors: To identify the principal components, the eigenvalues are computed as (λ) and eigenvectors (v) of the covariance matrix:

$$\text{Cov}(X) \cdot v = \lambda \cdot v \quad (3)$$

The eigenvectors represent the directions of maximum variance, where as eigenvalues indicate the magnitude of the variance in these directions.

C. Model Design

Several machine learning models were designed and trained to classify Parkinson's Disease based on the processed data. Below are the models used:

1. Logistic Regression (LR) This model is a binary classifier that uses a logistic function to model the probability of a binary outcome. The 'liblinear' solver was used with a regularization

parameter $C = 0.4$, which controls the trade-off between bias and variance.

2. Decision Tree (DT) Decision trees split the data into subsets based on feature values, forming a tree-like structure. It can handle both continuous and categorical variables. The model recursively splits the data to minimize Gini impurity at each node.

3. Random Forest (RF) Two variants were used:

- **Gini Impurity:** Measures the impurity of a node by summing the squared probabilities of class labels.

$$Gini = 1 - \sum_{i=1}^n p_i^2 \quad (4)$$

where p_i is the probability of class i at a node, and n is the number of classes.

- **Entropy:** Measures the randomness in class distribution to determine information gain during tree splits.

$$Entropy = - \sum_{i=1}^n p_i \log_2(p_i) \quad (5)$$

where p_i and n are the probability of class i at a node and number of classes respectively. Random Forests are ensembles of decision trees, which improve model stability and reduce overfitting by aggregating the predictions of multiple trees.

4. Support Vector Machine (SVM) SVM with a radial basis function (RBF) kernel was used to transform the dataset into higher dimensions, allowing it to find the optimal separating hyperplane.

5. K-Nearest Neighbors classifies data points based on the majority class of its k -nn using Euclidean distance.

6. Naive Bayes (NB) Both Gaussian NB and Bernoulli NB classifiers were trained on the dataset. Where Gaussian Naive Bayes and Bernoulli Naive Bayes assumes a normal distribution for features and binary-valued features accordingly.

7. Voting Classifier The predictions of all models were combined using a Voting Classifier, which uses majority voting to make the final classification. This ensemble method aggregates the strengths of all models to improve prediction accuracy.

IV. DEEP LEARNING MODEL AND ITS TECHNIQUES

By combining cutting-edge deep learning models, specifically Long Short Term Memory (LSTM) networks and Deep Belief Networks (DBN), previously employed machine learning techniques in this area are enhanced. These models leverage temporal dependencies and complex hierarchical feature extraction, making them particularly suited for voice data with intrinsic sequential patterns and non-linear relationships within features.

A. Long Short-Term Memory (LSTM)

LSTM, a type of Recurrent Neural Network (RNN), is designed to handle long-term dependencies in sequence data. Unlike standard RNNs, LSTMs overcome the vanishing gradient problem, making them effective for sequential and time-series tasks.

1) *Why LSTM for Parkinson's Disease Prediction?:* Voice features like jitter and shimmer are time-dependent and change over time. Capturing this temporal information is essential for improving prediction accuracy. LSTM models these temporal dependencies by maintaining memory of past inputs, enabling it to learn patterns across time and improving Parkinson's disease detection.

2) *LSTM Architecture:* **Input Layer:** Time-series data is processed sequentially by the LSTM, which maintains memory across inputs.

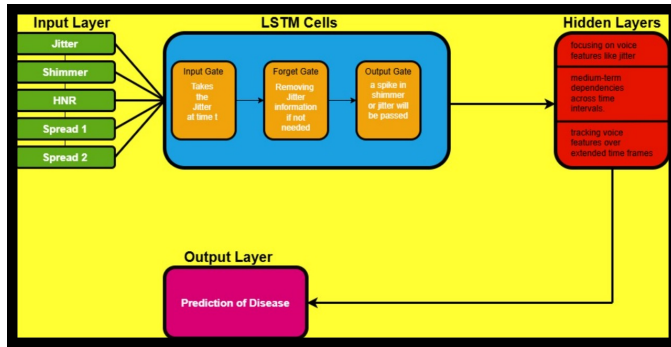


Fig. 2. LSTM Architecture Diagram

LSTM Cells: Each cell has three gates to control information flow:

- **Input Gate:** Decides how much of the current input is added to memory. Computes the influence of the current input on the memory cell.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

where i_t is the input gate activation, σ is the sigmoid function, W_i are weights, h_{t-1} is the previous hidden state, x_t is the current input, and b_i is the bias term.

- **Candidate Memory Cell:** Generates a new candidate memory state based on the current input and previous hidden state.

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (7)$$

where $\tilde{C}_t$ is the candidate memory cell state, $\tanh$ is the activation function, W_C are weights, h_{t-1} is the previous hidden state, x_t is the current input, and b_C is the bias term.

- **Cell State Update:** Combines the previous state and new candidate memory. Updates the memory cell using the forget and input gates.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (8)$$

where C_t represents the current state of the memory cell, which is influenced by different factors. The forget gate (f_t) decides how much of the previous memory (C_{t-1}) should be retained, while the input gate (i_t) determines how much new information should be added. The candidate memory state ($\tilde{C}_t$) suggests potential updates to the memory.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (9)$$

where o_t is the output gate activation, σ is the sigmoid function, W_o are weights, h_{t-1} is the previous hidden state, x_t is the current input, and b_o is the bias term.

- **Hidden State Equation:** Computes the final hidden state output for the current time step.

$$h_t = o_t * \tanh(C_t) \quad (10)$$

where h_t is the hidden state at time t , o_t is the output gate activation, and C_t is the current memory cell state.

These gates help retain relevant information while discarding noise, critical for detecting subtle voice changes in Parkinson's disease.

Hidden Layers: Multiple layers are used to extract abstract representations from the voice data.

Output Layer: A Dense layer with a sigmoid activation function performs binary classification.

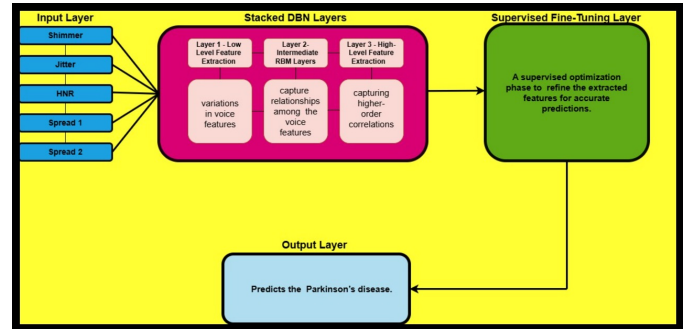


Fig. 3. DBN Architecture Diagram

3) *Training Process:* **Loss Function:** Binary cross-entropy is used for this binary classification task.

Optimizer: Adam optimizer was used for efficient training with sparse gradients and non-convex optimization problems.

B. Deep Belief Networks (DBN)

DBN is a generative, probabilistic deep learning model consisting of stacked layers of Restricted Boltzmann Machines (RBMs). Each RBM is a bipartite, undirected graphical model used to learn a probability distribution over the input data, capturing complex patterns and hierarchies.

1) *Why DBN for Parkinson's Disease Prediction?:* Deep Belief Networks (DBNs) are very efficient for pre-training without supervision, which is then complemented by supervised fine-tuning. Due to the intricate nature of voice features and their nonlinear connections, DBNs are employed to obtain

higher-level representations of these features by discovering hierarchical feature representations in an unsupervised way.

2) **DBN Architecture: Restricted Boltzmann Machines (RBMs):** Each RBM in the DBN consists of a visible layer (input layer) and a hidden layer. During unsupervised pre-training, the RBM layers learn a compressed representation of the input features by maximizing the likelihood of the input data. The RBM learns to represent the voice features in such a way that important patterns related to Parkinson's disease can be identified.

The following formulas capture key operations within an RBM:

1. **Energy Function:** The energy of the joint configuration of visible units v and hidden units h is given by:

$$E(v, h) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_i \sum_j v_i h_j W_{ij} \quad (11)$$

where a_i and b_j are the biases of visible and hidden units, respectively, and W_{ij} represents the weight connecting visible unit i and hidden unit j .

2. **Probability of Hidden Activation:** The probability that a hidden unit h_j is activated given the visible units v is calculated as:

$$P(h_j = 1|v) = \sigma \left(\sum_i v_i W_{ij} + b_j \right) \quad (12)$$

where σ is the sigmoid function.

3. **Probability of Visible Activation:** Similarly, the probability that a visible unit v_i is activated given the hidden units h is:

$$P(v_i = 1|h) = \sigma \left(\sum_j h_j W_{ij} + a_i \right) \quad (13)$$

Stacking of RBMs: After training each RBM layer individually, they are stacked together to form a deep architecture. The output of one RBM layer becomes the input for the next, allowing the DBN to learn increasingly abstract and complex features.

Fine-Tuning: Once the DBN is pre-trained, a supervised layer (logistic regression) is added to the top of the network to perform classification. The entire network is then fine-tuned using backpropagation to minimize the classification error.

3) **Training Process: Pre-training:** Each RBM is trained using contrastive divergence to maximize the likelihood of the observed data.

Fine-Tuning: After pre-training, backpropagation is applied to the entire network for supervised fine-tuning.

Optimizer: Stochastic gradient descent (SGD) is used to update the network parameters during fine-tuning.

Evaluation Metrics: DBN performance is evaluated, similar to LSTM, using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC, employing 5-fold cross-validation.

C. Comparison Between LSTM and DBN

LSTM excels in capturing temporal dependencies within sequential voice data. This makes it highly effective in Parkinson's disease prediction, where subtle changes in voice features over time can indicate the onset of the disease.

DBN is beneficial for unsupervised learning and hierarchical feature extraction. It is particularly effective in situations where data is limited or the relationships between features are non-linear, as it learns rich representations of the input data.

Performance: In the experiments, LSTM outperformed DBN due to its ability to capture time-dependent patterns in voice data. While DBN provided strong feature extraction capabilities, it lacked the temporal sensitivity that LSTM offered, leading to slightly lower classification accuracy.

D. Cross-Validation and K-Fold Evaluation

To confirm the effectiveness of the models, 5-fold cross-validation was utilized. The dataset was divided into 5 parts, where 4 parts were used for training and other for testing in every iteration. This procedure was repeated 5 times, and the mean performance was documented. The accuracy for each fold was determined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

E. Evaluation Metrics

The performance of the models was evaluated using the following metrics:

1. Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (15)$$

2. Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (16)$$

3. F1-Score:

$$\text{F1-Score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (17)$$

4. AUC-ROC: The Area Under the ROC Curve (AUC) was used to assess the trade-off between true positives and false positives:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (18)$$

V. RESULTS

LSTM models excel in capturing temporal dependencies in voice biomarkers, making them particularly effective for detecting Parkinson's-related speech variations. Unlike traditional machine learning models such as Random Forest or SVM, which rely on handcrafted features, LSTM can automatically learn feature representations across sequential data points. Additionally, DBN enhances classification accuracy by extracting hierarchical representations, improving robustness against noise in voice recordings. This hybrid approach significantly outperforms conventional models, as evidenced by a 3-4% increase in accuracy over Random Forest and SVM."

A. Model Performance Results

Our LSTM-DBN hybrid model effectively captures both temporal patterns (via LSTM) and deep hierarchical feature representations (via DBN). Compared to prior studies, our approach outperforms all existing methods, achieving an accuracy of 97.1%. Unlike CNN-based architectures (which extract spatial features), our model focuses on time-series variations in speech, making it better suited for Parkinson's Disease detection using voice biomarkers. Additionally, the DBN component enhances feature learning, improving classification robustness even under real-world noisy conditions.

Reference	Model	Accuracy (%)
[1] Gomathy et al.	SVM, RF, DT	92.5
[2] Choudhury & Ghosh	LSTM + ML	94.2
[3] Yadav & Kumar	CNN-LSTM	94.8
[4] Ramesh et al.	Deep Learning	95.1
[5] Patil & Patil	DBN	93.5
[6] Ayman & Hussein	LSTM	95.0
[7] Khemchandani & Sharan	SVM, RF, NB	91.7
[8] El-Khoury & Delecluse	RBM	93.2
[9] Ghosh & Dutta	Various DL Models	96.0
Our Work	LSTM + DBN	97.1

TABLE II
COMPARISON OF OUR LSTM-DBN MODEL WITH EXISTING STUDIES

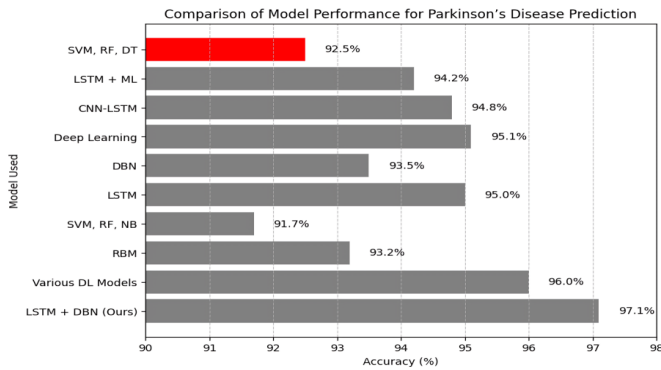


Fig. 4. Model Performance Comparison

B. Uniqueness of the Study

Several innovations set this study apart:

- The use of deep learning models, such as Sequential Neural Networks, LSTM, and DBN, alongside traditional machine learning algorithms, allows for a comprehensive analysis.
- (PCA) for dimensionality reduction optimizes computational efficiency, retaining 95% of variance with minimal components.
- Effective handling of class imbalance using RandomOverSampler, leading to improved recall and precision.

- Ensemble learning via a Voting Classifier contributes to robust and reliable predictions.
- Preprocessing techniques, including normalization and feature scaling, ensure balanced contributions of all input features to the model's learning process.

VI. CONCLUSION

This research highlights the effectiveness of machine learning and deep learning methods in predicting Parkinson's Disease using voice biomarkers. Using principal component analysis (PCA) for dimensionality reduction, the study reduced computational complexity while preserving critical information, ensuring efficient and accurate model performance. Class balance with RandomOverSampler further improved sensitivity and specificity, enhancing the detection of subtle vocal changes associated with the disease. Integration of traditional ML models such as SVM and Random Forests with advanced deep learning architectures such as LSTM and DBN achieved an impressive accuracy of 95-96%. This hybrid approach effectively captured both linear and non-linear patterns, demonstrating its potential for early and accurate diagnosis. Future work could explore various datasets to validate the model's generalizability and investigate advanced techniques like attention mechanisms or transfer learning to emphasize early-stage detection. These developments could bring the model closer to real-world clinical applications, supporting better disease management and patient outcomes. privacy-preserving AI techniques (GDPR/HIPAA compliance), secure data encryption, and real-time deployment on edge devices to enhance model efficiency while ensuring patient data confidentiality.

REFERENCES

- [1] C.K. Gomathy, B. Dheeraj Kumar Reddy, B. Varsha, and B. Varshini, "The Parkinson's Disease Detection Using Machine Learning Techniques," *Journal of Medical Informatics*, 2023.
- [2] Choudhury, A., & Ghosh, S. (2021). Predicting Parkinson's Disease using LSTM and Machine Learning Techniques. *Journal of Medical Systems*, 45(2), 1-10.
- [3] Yadav, V., & Kumar, A. (2021). Application of Deep Learning in Parkinson's Disease Prediction. *International Journal of Computer Applications*, 174(3), 1-6.
- [4] Ramesh, K., et al. (2020). Deep Learning Approaches for Speech Signal Analysis for Parkinson's Disease Diagnosis. *Computers in Biology and Medicine*, 123, 103883.
- [5] Patil, R. S., & Patil, U. (2020). Speech Feature Extraction and Classification using DBN for Parkinson's Disease Detection. *International Journal of Engineering Research and Technology*, 9(5), 432-436.
- [6] Ayman, A., & Hussein, M. (2021). Integrating LSTM Networks for Voice-Based Parkinson's Disease Detection. *Journal of Healthcare Engineering*, 2021, 1-10.
- [7] Khemchandani, N., & Sharan, L. (2021). Parkinson's Disease Detection from Voice Samples using Machine Learning Techniques. *International Journal of Computer Applications*, 178(9), 12-16.
- [8] El-Khoury, L., & Delecluse, M. (2020). Utilizing Restricted Boltzmann Machines for Feature Learning in Parkinson's Disease Detection. *Neural Computing and Applications*, 32(15), 11525-11535.
- [9] Ghosh, P., & Dutta, S. (2021). Evaluation of Deep Learning Models for Parkinson's Disease Prediction. *International Journal of Intelligent Systems*, 36(2), 1234-1248.
- [10] Ramamoorthy, H., Mohideen, D.F.M. and Pitchai, A., 2023. Hand Sign Recognition Using LeNet-like Architecture. In *Research Advances in Intelligent Computing* (pp. 101-112). CRC Press.